

LISA NGUYEN

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EDUCATION

University of Michigan

PhD in Applied Physics and Scientific Computing

Topic: Spatio-Temporal Dynamics and Causal Drivers of Hydro-climatic Extremes Under Climate Change

Expected Graduation: Dec 2026

University of California Berkeley

B.A. in Physics

B.A. in Pure Mathematics

Graduation: Dec 2019

AWARDS/FELLOWSHIPS/PROGRAMS

- Rackham Predoctoral Fellowship, 2026-27 academic year (≈ 85 selected campus-wide)
- University of Michigan student delegate for COP30 (1 of 8 candidates selected campus-wide)
- Thriving Earth Exchange Community Science Fellowship
- Future of Research in Climate, Earth, and Energy (FORCEE) DoE Internship
- Rackham Merit Fellowship
- Joint PhD program in Scientific Computing, Michigan Institute for Computational Discovery and Engineering (MICDE)

RESEARCH EXPERIENCE

Hydrological Extremes with HEIDA Lab

Advisor: Mohammed Ombadi

University of Michigan

2024 - present

- **Global Drought Synchronization and Breadbasket Exposure** (ongoing): Investigated how warming changed the risk of synchronized droughts across major crop-producing regions, using CMIP6 projections. Investigated the relative roles of precipitation deficits and atmospheric evaporative demand in drought synchronicity by comparing SPI- and SPEI-based patterns.
- **Intercomparison of Drought Clustering Methods** (ongoing): Compared methods for identifying and tracking drought events in space and time; assessed how method parameterizations affected event duration, spatial extent, intensity, and migration to distinguish robust drought characteristics from methodological differences.
- **Drought Synchronicity over the Contiguous United States** (Apr-Dec 2025): Characterized the spatio-temporal synchronization of meteorological droughts over the contiguous United States. Examined links between drought synchronicity and large-scale climate variability using event coincidence analysis, wavelets, and Random Forest/SHAP.
- **Record-Breaking Extreme Precipitation Events** (Apr 2024-May 2025): Assessed how global warming changed record-breaking precipitation events from a hazard/exposure/vulnerability perspective, combining CMIP6 projections with socioeconomic data.

Urban Heat Resilience Strategies AGU: Thriving Earth Exchange

Community Science Fellow

Boynton Beach, FL

2025 - present

- **Hot Zones: Urban Heat Resilience and Cooling-Center Planning:** Led a project to create a detailed roadmap for implementing cooling centers for Boynton Beach. Oversaw a nine-person team to research the risk of urban heat and prioritize objectives between researchers, municipal leaders, and non-profits.

SAI modeling with FORCEE Research Internship

Supervisor: Lauren Wheeler

Sandia National Laboratories

May 2023 - Oct 2023

- **Stratospheric Aerosol Injection Modeling:** Investigated temperature control through stratospheric aerosol injection by validating an improved E3SMv2 aerosol chemistry as well as implementing a proportional-integral feedback controller to adjust injection rates against three temperature targets.

Stratospheric Dynamics with Atmospheric Dynamics Modeling Group

Advisor: Christiane Jablonowski

University of Michigan

May 2022 - Oct 2023

- **Stratospheric Circulation and Transport in E3SMv2:** Assessed how accurately the model represented tropical stratospheric circulation and transport using Transformed Eulerian Means to investigate the Brewer Dobson Cycle. Identified a dry bias and overly rapid upward propagation of water-vapor anomalies.

Power Spectra of Precipitation with Brian Arbic Group

Advisor: Brian Arbic

University of Michigan

Aug 2021 - Jan 2022

- **Frequency Analysis of High-Intensity Precipitation:** Evaluated how well high-resolution climate models captured rainfall intensity from minutes to decades by comparing precipitation power spectra with high-frequency NOAA rain-gauge observations.

PEER-REVIEWED PUBLICATIONS

- **Nguyen, L.** & Ombadi, M. (2026) *Characterizing Patterns of Drought Synchronicity in the Contiguous United States*. Water Resources Research. <https://doi.org/10.1029/2025WR041240>
- **Nguyen, L.**, Todorovic, L., & Ombadi, M. (2026) *Anthropogenic Global Warming Increases the Risk of Record-breaking Extreme Precipitation Events in low-income Countries*. Nature Communications Earth & Environment. <https://doi.org/10.1038/s43247-026-03649-y>
- Ombadi, M., **Nguyen, L.**, Gaur, S., Gronewold, A., & Reich, P. (2026) *Increased Dependency on Extreme Precipitation in a Warmer Climate*. Water Resources Research. <https://doi.org/10.1029/2025WR042627>

MANUSCRIPTS IN PREPARATION

- **Nguyen, L.** & Ombadi, M. *Intercomparison of Object-Oriented Drought Clustering Methods*. In preparation.
- **Nguyen, L.** & Ombadi, M. *Drought Synchronization in a Warming World and Impacts on Breadbasket Regions*. In preparation.

POSTERS/PRESENTATIONS

- **Nguyen, L. et al.** (AGU 2026) *Bridging Earth Observation and Community Knowledge: A Heat Resilience Case Study in Boynton Beach, Florida*, American Geophysical Union (AGU) 2026, Abstract 2095314: GH001. Accepted.
- **Nguyen, L.** (March 17, 2026) Invited speaker for *Student Stories from COP30: COP30 Debrief with Environmental Policy Association*, University of Michigan Environmental Policy Association
- **Nguyen, L.** & Ombadi, M. (January 29, 2026), *Quantifying Global Synchronization of Droughts in Breadbasket Regions Under a Warming Climate*, American Meteorological Society (AMS) Annual Meeting, Abstract 469617
- **Nguyen, L.** & Ombadi, M. (December 19, 2025), *Characterizing Drought Synchronicity Across the Contiguous United States (1980–2021)*, American Geophysical Union (AGU) 2025, Abstract 1969758: NH51E-0228
- **Nguyen, L.**, Li, Q., & Ombadi, M. (December 18, 2025), *Decoding Drought Dynamics: A Multi-Method Intercomparison for Event Characterization Across Spatio-Temporal Scales*, American Geophysical Union (AGU) 2025, Abstract 1969758: H41M-1332
- Todorovic, L., **Nguyen, L.**, & Ombadi, M. (December 17, 2025), *Disproportionate Increase of Unprecedented Rainfall Events in Low Income Countries*, American Geophysical Union (AGU) 2025, Abstract A31E-2108
- Gaur, S., **Nguyen, L.**, Gronewold, A., Reich, P., & Ombadi, M. (December 15, 2025), *Changes in Extreme and Annual Precipitation under Global Warming*, American Geophysical Union (AGU) 2025, Abstract GC11G-0662
- Zhong, Y., Li, Q., **Nguyen, L.**, & Ombadi, M. (December 18, 2025), *Examining the Impact of Drought on Vegetation in the Contiguous United States*, American Geophysical Union (AGU) 2025, Abstract B41I-1970
- **Nguyen, L.** (August 13, 2025) Invited speaker for *Webinar: Science with Purpose: How Scientists Can Help Empower Community Solutions*, American Geophysical Union, Thriving Earth Exchange
- **Nguyen, L.**, Ombadi, M. (2024), *Quantifying Global Event Synchronization of Droughts in a Warming Climate*, American Geophysical Union (AGU) 2024, Abstract 1620436: GC33G-0234
- Jablonowski, C., Hollowed, J., **Nguyen, L.**, Hughes, O., Ehrmann, T., Wagman, B., & Hillman, B. (2023), *A Closer Look at E3SMs Stratosphere: Circulation Biases, Transport, Tracers*, DoE CLDERA All-Hands Meeting, Albuquerque, NM, USA, Oct 17-19, 2023
- Jablonowski, C., Hollowed, J., **Nguyen, L.**, Ehrmann, T., Wagman, B., & Hillman, B. (2023), *Tropical Stratosphere-Troposphere Interactions in Selected CMIP6 Models*, Joint SPARC DynVar - SNAP Meeting, Munich, Germany, Oct. 9-13, 2023
- Jablonowski, C., **Nguyen, L.**, Kinnison, D., & the CLDERA project team (2023), *Characteristics of the stratospheric tropical circulation in DoEs Energy Exascale Earth System Model E3SMv2*, CESM Workshop 2023, Boulder, CO, USA, June 12-14, 2023
- Jablonowski, C., **Nguyen, L.**, Golaz, J.-C., Rosenbloom, N., & Meehl, G. A. (2023), *Characteristics of the Stratospheric Tropical Circulation of the Energy Exascale Earth System Model E3SMv2*, European Geosciences Union (EGU) 2023, Abstract EGU23-15173

COMMUNITY LEADERSHIP

- AGU: Thriving Earth Exchange** **Detroit, MI**
Volunteer Community Scientist Aug 2024 - Dec 2025
• **Illegal Dumping Mitigation and Environmental Education:** Supported waste mitigation efforts at Scott Memorial Church through research, recycling workshops, and coordination with local muralists.
- Climate Blue at University of Michigan** **Ann Arbor, MI**
COP30 Student Delegate May 2025 - present
• Represented Umich as a Climate Blue delegate to the UNFCCC COP 30, producing accessible briefings and outreach materials to connect global climate negotiations with students, researchers, and local communities.
- University of Michigan: CLaSP GUSTO** **Ann Arbor, MI**
DEI Officer and Volunteer Graduate Mentor Aug 2024 - present
• Organized DEI related activities and mentorship within the department.
• Volunteered to mentor graduate students interested as part of the GUSTO mentorship program.
- Ypsilanti Math Corp at University of Michigan** **Ann Arbor, MI**
Course Instructor Summer 2024
Math Corp Activities/HW Volunteer May 2019 - Dec 2024
• Developed and implemented "Ice Cream" class curriculum on graph theory for 8th graders and high school to teach fun topics in math during the summer camp.
• Co-organized the activities room to increase awareness and engagement with mathematics for Ypsilanti children grades 7-9 at the Ypsilanti Math Corp summer camp and Super Saturdays.
- FATE Give Merit Program** **Ann Arbor, MI**
Mentor Sep 2022 - Dec 2024
• Mentored high school students weekly in Detroit, working on strengthening their professional skills.
- University of Michigan: Applied Physics Department** **Ann Arbor, MI**
Volunteer Graduate Mentor Aug 2022 - present
• Mentored incoming graduate student, Shriya Sinha. Set up weekly HW sessions and monthly dinners.
- Physics Diversity, Equity, and Inclusion Committee** **Ann Arbor, MI**
Outreach Committee Volunteer Aug 2021 - present
• Volunteered for outreach opportunities with the local communities. (WISE middle school demos, etc)
- Michigan Math Circle** **Ann Arbor, MI**
Volunteer Assistant Aug 2022 - May 2023
• Guided elementary to high school students in engaging and complex math topics.
• Gave interactive lectures and activities on accessible, but interesting math problems
- Applied Physics Student Council** **Ann Arbor, MI**
Authorized Signer Aug 2022 - present
• Responsible for activities hosted by the Applied Physics Department's graduate student council.
- University of Michigan: College of Engineering** **Ann Arbor, MI**
Volunteer Graduate Mentor Aug 2021 - Aug 2022
• Volunteered to mentor undergraduate students interested in climate and space. Set up regular HW sessions and talks to discuss various aspects of academic and social life.
- Cal Women's Rugby Team** **Berkeley, CA**
Academic Officer/(Player Position: Prop) Aug 2018 - Jun 2019
• Monitored and created individual academic plans, with an emphasis on fostering time management skills.
• Scheduled and organized study sessions and events to promote an academic lifestyle.
- Mathematics Undergraduate Student Associations (MUSA)** **Berkeley, CA**
Berkeley Undergraduate Mathmentor Program (BUMP) Mentor Aug 2018 - Jun 2019
• Each semester, mentored two underclass students interested in mathematics in order to foster enthusiasm for continuing math education. Met twice a week to talk about school/hobbies/life.

SKILLS

- **Computing:** Python (xarray, NumPy, pandas, scikit-learn, PyTorch, Dask, Matplotlib), Bash, Git, LaTeX, HPC batch environments, and terabyte-scale NetCDF workflows.
- **Methods and models:** Event Coincidence Analysis, clustering, Random Forest/SHAP, wavelet and spectral analysis; E3SMv2, CESM, WRF; CMIP6, HighResMIP, reanalysis, and station observations.
- **Additional tools:** AWS, Power BI; working knowledge of MATLAB, Fortran, C++, R, and SQL.

TEACHING

- **Physics 141: Elementary Lab** (GSI) - Winter 2024

WORK EXPERIENCE

RSAE Labs

Bookkeeper

Tampa Bay, FL

Jul 2021 - present

- In charge of payroll, accounts receivable, accounts payable, and financial reports. Streamlined several administrative processes, to facilitate transparency.

Omnium

Data Scientist

San Francisco, CA

Jul 2019 - Aug 2021

- Classification of SKU rationalization using various clustering models.
- ETL on Consumer Package Goods (CPG) consumption data to derive key insights for client business intelligence.
- Optimized pivotal client business metrics (both top and bottom line) given budget constraints by building a heuristic algorithm to a Mixed Integer Non-Linear programming problem.
- Restructured and reformatted internal data capture processes and automated lower level analysis by establishing a coherent data model and pipeline using AWS and Power BI.
- Significantly streamlined and implemented intern training to create a systematic transition from training to internal and client facing company projects. Served as a project mentor for 8 different interns.

Sandia National Laboratories

Nuclear Radiation and Detection Intern (NSSC affiliate)

Livermore, CA

Aug 2018 - Jun 2019

- Oversaw and implemented a methodical study of neutron production in high-Z materials (lead and tungsten) on a per-muon basis, with the goal of calculating the neutron multiplicity distribution for both spallation and capture channels. Optimized the data acquisition system, independently researched and ran experiments.

MISCILLANEOUS

- I am PADI advanced scuba certified and always looking for partners to explore the waters with!
- I have my Accelerated Free Fall (AFF) license to skydive
- I am an avid baker, board gamer